

Saiyang Zhang

Department of Physics, University of Texas at Austin · 2515 Speedway, Austin, TX 78712
szhangphys@utexas.edu | sylvanzsy.github.io | Google Scholar | GitHub | ORCID:
0000-0003-1541-177X

Research Interests

Dark matter (primordial black holes, WIMPs, warm DM, SIDM); early universe cosmology; formation of the first stars and galaxies; supermassive black holes and AGN; JWST observations; large-scale structure; cosmological N-body/hydrodynamical simulations; Bayesian inference; scientific machine learning

Upcoming Appointments

CITA Postdoctoral Fellow — Canadian Institute for Theoretical Astrophysics	Fall 2026
Arts & Science Postdoctoral Fellow — University of Toronto	Fall 2026

Education

Ph.D. in Physics University of Texas at Austin Advisor: Volker Bromm Thesis: <i>Imprints of the Primordial Black Holes over Cosmic History</i>	Aug 2020 – Aug 2026 (exp.) Austin, TX
M.A. in Physics University of Texas at Austin Advisor: Katherine Freese Thesis: <i>Detecting Supermassive Dark Stars with the Roman Space Telescope</i>	Aug 2020 – Dec 2024 Austin, TX
B.A. in Astronomy/Physics with Honors & Applied Mathematics (<i>Magna cum laude</i>) Colgate University Advisor: Cosmin Ilie Thesis: <i>Multi-scatter Capture of Super Heavy Dark Matter by the First Stars</i>	Aug 2015 – May 2019 Hamilton, NY

Research Experience

2022–present	Graduate Research Assistant Advisor: Volker Bromm Thesis: <i>Imprints of Primordial Black Holes over Cosmic History. N-body/SPH simulations (GIZMO/C++), analytical modeling, Bayesian inference vs. JWST. Co-I: JWST DD Program Cycle 4, DD 9433.</i>	UT Austin
2025	Research Student, DREAMS Project Supervisor: Arya Farahi WDM constraints via ML inference on 1,024 hydrodynamical simulations	UT Austin
2021–2024	Graduate Research Assistant Advisor: Katherine Freese Detectability of supermassive dark stars with Roman Space Telescope and JWST (PANDEIA + TLUSTY).	UT Austin
2018–2020	Research Assistant Advisor: Cosmin Ilie Multiscatter DM capture by Pop III stars and neutron stars; constraints on DM–baryon cross sections.	Colgate University

Publications

* = first or co-first author · h-index: 11 · 416 total citations (NASA/ADS) · [Full list on NASA/ADS](#)

Peer-Reviewed Journals

1. Juodžbalis, I., ..., **Zhang, S.**, ..., et al. (2026). *A direct black hole mass measurement in a Little Red Dot at the Epoch of Reionization*. Accepted by *Nature*. [[arXiv:2508.21748](#)]
2. Maiolino, R., ..., **Zhang, S.**, ..., et al. (2026). *A black hole in a near-pristine galaxy 700 million years after the Big Bang*. *MNRAS*, 548(1), staf2109. [[arXiv:2505.22567](#)]
3. ***Zhang, S.**, Liu, B., Bromm, V., & Kühnel, F. (2026). *Primordial Black Holes as Seeds for Extremely Overmassive AGN Observed by JWST*. *ApJL*, 1000(1), L19. [[arXiv:2512.14066](#)]
4. ***Zhang, S.**, Liu, B., & Bromm, V. (2025). *A Novel Formation Channel for Supermassive Black Hole Binaries in the Early Universe via Primordial Black Holes*. *ApJ*, 992(1), 136. [[arXiv:2508.00774](#)]
5. ***Zhang, S.**, Liu, B., Bromm, V., Jeon, J., Boylan-Kolchin, M., & Kühnel, F. (2025). *How do Massive Primordial Black Holes Impact the Formation of the First Stars and Galaxies*. *ApJ*, 987(2), 185. [[arXiv:2503.17585](#)]
6. ***Zhang, S.**, Bromm, V., & Liu, B. (2024). *How do Primordial Black Holes change the Halo Mass Function and Structure*. *ApJ*, 975(1), 139. [[arXiv:2405.11381](#)]
7. ***Zhang, S.**, Ilie, C., & Freese, K. (2024). *Detectability of Supermassive Dark Stars with the Roman Space Telescope*. *ApJ*, 965(2), 121. [[arXiv:2306.11606](#)]
8. ***Zhang, S.**, Liu, B., & Bromm, V. (2023). *Distinguishing the impact and signature of black holes from different origins in early cosmic history*. *MNRAS*, 528, 180. [[arXiv:2310.01763](#)]
9. Liu, B., **Zhang, S.**, & Bromm, V. (2022). *Effects of stellar-mass primordial black holes on first star formation*. *MNRAS*, 514, 2376. (see also *Nature Astronomy*, 6, 881, 2022.) [[arXiv:2204.06330](#)]
10. Ilie, C., Levy, C., Pilawa, J., & **Zhang, S.** (2020). *Constraining dark matter properties with the first generation of stars*. *Phys. Rev. D*, 104, 123031. [[arXiv:2009.11474](#)]
11. Ilie, C., Pilawa, J., & **Zhang, S.** (2020). *Comment on “Multiscatter stellar capture of dark matter”*. *Phys. Rev. D*, 102, 048301. [[arXiv:2005.05946](#)]
12. *Ilie, C., & **Zhang, S.** (2019). *Multiscatter capture of superheavy dark matter by Pop III stars*. *JCAP*, 2019(12), 051. [[arXiv:1908.02700](#)]
13. Weaver, Z. R., Balonek, T. J., ..., & **Zhang, S.** (2019). *The 2016 June Optical and Gamma-Ray Outburst and Optical Microvariability of the Blazar 3C 454.3*. *ApJ*, 875(1), 15. [[arXiv:1903.04587](#)]

Preprints & Conference Papers

14. *Jeon, J., Jeong, T., **Zhang, S.**, & Bromm, V. (2026). *What is Powering the Enigmatic He II Emitter Hebe: The First Stars or Black Holes?* Submitted to *ApJ*. [[arXiv:2604.19075](#)]
15. Ivey, L., ..., **Zhang, S.**, ..., et al. (2026). *The Cliff: A Metal-Poor Little Red Dot Hosting an Overmassive Black Hole at $z = 3.55$* . Submitted to *MNRAS*. [[arXiv:2604.09177](#)]
16. Song, Z., ..., **Zhang, S.**, ..., et al. (2025). *Evaluating Large Language Models in Scientific Discovery*. Submitted to *Nature Computational Science*. [[arXiv:2512.15567](#)]
17. Übler, H., ..., **Zhang, S.**, ..., et al. (2025). *BlackTHUNDER: evidence for three massive black holes in a $z \sim 5$ galaxy*. Submitted to *A&A*. [[arXiv:2509.21575](#)]

18. Ilie, C., & **Zhang, S.** (2020). *Erratum: Multiscatter capture of superheavy dark matter by Pop III stars.* JCAP, 2020(10), E01.
19. Ilie, C., Pilawa, J., & **Zhang, S.** (2020). *Probing below the neutrino floor with the first generation of stars.* ArXiv preprint. [[arXiv:2009.11478](https://arxiv.org/abs/2009.11478)]
20. Galvez, E. J., Dutta, I., & **Zhang, S.** (2018). *Möbius Polarization of Non-Collinear Poincaré Superpositions.* Latin America Optics and Photonics Conference, Tu3D-2.

Astronomer's Telegrams

- ATel 1. Balonek, T. J., . . . , **Zhang, S.**, . . . *Yet Another Extraordinary Optical Flare in the Blazar CTA 102.* ATel 9808 (2016).
- ATel 2. Chapman, K. J., . . . , **Zhang, S.**, . . . *A Spectacular, Unprecedented Optical Flare in the Blazar CTA 102.* ATel 9756 (2016).
- ATel 3. Balonek, T. J., **Zhang, S.**, et al. *Blazar CTA 102 Reaches Historic Optical Maximum.* ATel 9732 (2016).
- ATel 4. Balonek, T. J., . . . , **Zhang, S.**, . . . *Optical Observations of Correlated Outburst Activity in BL Lacertae.* ATel 9600 (2016).
- ATel 5. Balonek, T. J., . . . , **Zhang, S.**, . . . *Detection of optical outburst activity in OT 081 (PKS 1749+096).* ATel 9259 (2016).
- ATel 6. Balonek, T. J., . . . , **Zhang, S.**, . . . *A Dramatic Optical Flare and Microvariability in the Blazar 3C 454.3.* ATel 9178 (2016).

Talks & Presentations

Invited Talks (●) and Contributed Talks / Seminars (★)

- Dec. 2025 ★ Group Meeting Talk, Peking University, Beijing, China
- Dec. 2025 ★ Texas Symposium on Relativistic Astrophysics, ASU, Tempe, AZ
- Oct. 2025 ★ Extragalactic Discussion Group, University of Hawai'i at Mānoa, HI
- Sep. 2025 ★ Thunch Talk, Princeton University, Princeton, NJ
- Sep. 2025 ★ Galaxy Lunch, Yale University, New Haven, CT
- Sep. 2025 ★ PACMAN Seminar, Flatiron Institute, New York City, NY
- Sep. 2025 ★ Seminar, Columbia University, New York City, NY
- Aug. 2025 ★ Santa Cruz Galaxy Workshop ([video](#)), UC Santa Cruz, CA
- Apr. 2025 ★ ITC Luncheon Talk ([video](#)), Harvard Center for Astrophysics, Boston, MA
- Apr. 2025 ★ MAT Seminar, MIT Kavli Institute, Boston, MA
- Apr. 2025 ★ Group Meeting Talk, Carnegie Observatories, Pasadena, CA
- Dec. 2024 ★ Group Meeting Talk, Tsinghua University, Beijing, China
- Nov. 2024 ★ CFC-CCA Workshop, University of Texas, Austin, TX
- Oct. 2024 ● TEPAPP Seminar (*invited*), University of California, Los Angeles, CA
- Oct. 2024 ★ Astrophysics Seminar, University of California, Riverside, CA
- Jul. 2024 ● Seventeenth Marcel Grossmann Meeting (*invited*), Bologna, Italy
- Jul. 2024 ★ International Workshop on the Identification of Dark Matter, Italy
- Apr. 2024 ● Physics Seminar (*invited*), Colgate University, Hamilton, NY
- Apr. 2022 ★ APS April Meeting, New York City, NY
- Oct. 2016 ★ Keck Northeast Astronomy Consortium, Wesleyan University, CT

Posters

- May 2025 CFC2025, University of Texas, Austin, TX
 Mar. 2025 Dark Matter 2025, UCLA, Los Angeles, CA
 May 2024 First Stars VII, Flatiron Institute, New York, NY
 Dec. 2023 Texas Symposium on Relativistic Astrophysics, SJTU, Shanghai, China
 Jun. 2019 Symposium in Honor of the Legacy of Vera Rubin, Georgetown University, DC
 Mar. 2019 Rochester Symposium for Physics Students (SPS Regional), U of Rochester, NY

Grants & Proposals

- 2025 **JWST Director's Discretionary Program**, Co-I, Cycle 4, DD 9433
Black hole primacy — do supermassive black holes really predate galaxies?
 2024 **TACC HPC Allocation**, Co-I, AST23026 (30,000 node-hours)
The First Stars and Galaxies

Honors & Awards

- 2026 **CITA Postdoctoral Fellowship** — Canadian Institute for Theoretical Astrophysics, University of Toronto
 2026 **Arts & Science Postdoctoral Fellowship** — University of Toronto
 2025 **Dissertation Writing Fellowship** — UT Austin (\$24,000; competitive recognition of achievement)
 2024 **Robert S. Davis and Lyell P. Davis Scholarship Fund** — UT Austin (\$1,000; merit-based)
 2024–26 **Graduate School Professional Development Award** — UT Austin (\$2,250)
 2022 **APS DAP Student/Postdoc Travel Grant** — American Physical Society (\$600)
 2019 **Joseph C. Amato & Anthony F. Aveni Award** (Excellence in Thesis) — Colgate University
 2016–19 **Dean's Award with Excellence** (merit-based) — Colgate University
 2016 **Edwin Foster Kingsbury Prize & Sisson Mathematics Prize** (\$500) — Colgate University

Service, Mentoring & Teaching

Professional Service

- Referee: *Astronomical Journal (AJ)*, *Physical Review Letters (PRL)* 2025–present
- External Examiner, Honors Defense, Colgate University 2025–present
- Member: American Astronomical Society (AAS), American Physical Society (APS)

Mentoring

- Ken Nordhauser — Research Mentee, Visiting Student, UT Austin
- Kaitlyn Chang & Kyara Michelle — Directed Reading Program Mentees, UT Austin
- Jillian Paulin — Research Mentee (Dark Stars); now PhD student, Univ. of Pennsylvania

Teaching

- Grading Assistant, UT Austin: PHY352K (Fall 2025), PHY362K (Spring 2024), PHY373 (Fall 2023), PHY352L (Spring 2022)
- Lab Teaching Assistant, UT Austin: PHY 105M/N, 102N (2020–2025)

Technical Skills

Simulation codes	GIZMO (N-body/SPH, extended), MUSIC, ROCKSTAR, PANDEIA, TLUSTY
Analysis & ML	Python (NumPy, SciPy, Pandas, Astropy, Matplotlib), PyTorch, scikit-learn, YT, COLOSSUS; Bayesian inference; simulation-based inference (SBI)
Systems & HPC	C/C++, Fortran, Bash; MPI parallelization; SLURM; TACC Stampede3/Frontera
Other languages	MATLAB, Mathematica, R
Tools	Git/GitHub, L ^A T _E X, Jupyter, VS Code
Languages	Chinese (native), English (fluent), Japanese (elementary)

References

Prof. Volker Bromm

PhD Advisor

Jane and Roland Blumberg Centennial Professor
Department of Astronomy
University of Texas at Austin
110 Inner Campus Dr, Austin, TX 78712
vbromm@astro.as.utexas.edu
+1 (512) 689-0741

Prof. Mike Boylan-Kolchin

Collaborator & Weekly Group Meeting

Samuel T. and Fern Yanagisawa
Regents Professor of Astronomy
University of Texas at Austin
110 Inner Campus Dr, Austin, TX 78712
mbk@astro.as.utexas.edu
+1 (512) 471-3343

Prof. Arya Farahi

Research Supervisor (DREAMS Project)

Assistant Professor, Statistics & Data Sciences
CosmicAI Institute (NSF-Simons)
University of Texas at Austin
110 Inner Campus Dr, Austin, TX 78712
arya.farahi@austin.utexas.edu

Dr. Florian Kühnel

Collaborator

Senior Research Fellow
Max-Planck-Institut für Physik
Boltzmannstr. 8
85748 Garching, Germany
Florian.Kuehnel@physik.uni-muenchen.de

Prof. Pratika Dayal

Incoming CITA Host & Collaborator

Galaxy formation, Epoch of Reionization, dark matter

Professor, Canadian Institute for Theoretical Astrophysics (CITA)
University of Toronto, Toronto, ON, Canada
dayal@cita.utoronto.ca pratika24.wixsite.com/pratika-dayal

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